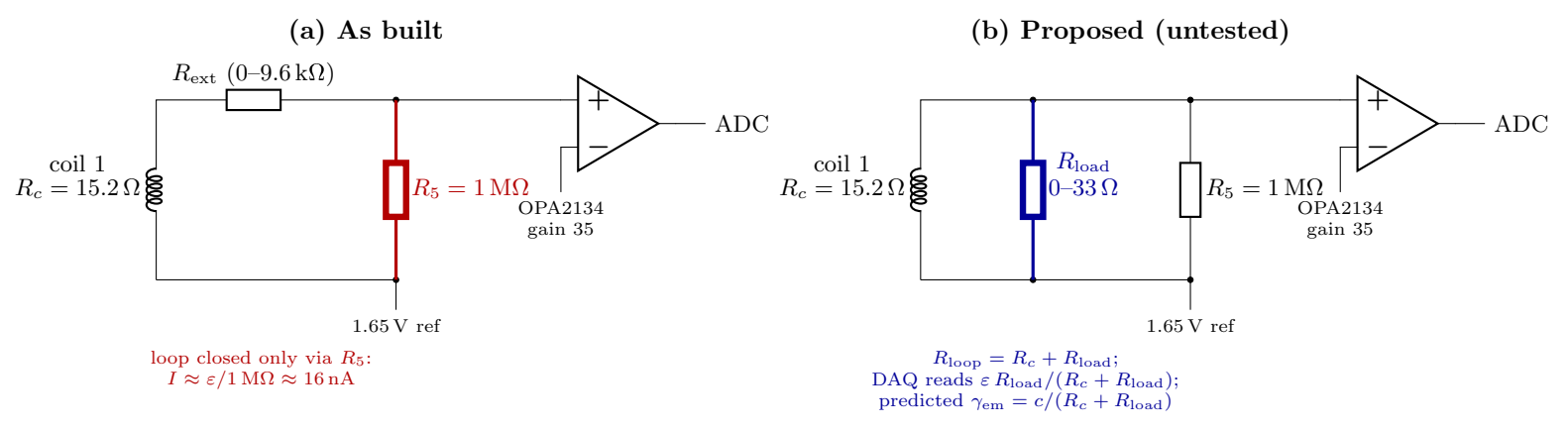

(a) As built
R_{ext} (0–9.6 kΩ)
coil 1
$R_c = 15.2\,\Omega$
$R_5 = 1\,\text{M}\Omega$
ADC
OPA2134
gain 35
1.65 V ref
loop closed only via R_5:
$I \approx \varepsilon/1\,\text{M}\Omega \approx 16\,\text{nA}$
(b) Proposed (untested)
coil 1
$R_c = 15.2\,\Omega$
R_{load}
0–33 Ω
$R_5 = 1\,\text{M}\Omega$
ADC
OPA2134
gain 35
1.65 V ref
$R_{\text{loop}} = R_c + R_{\text{load}}$;
DAQ reads $\varepsilon\, R_{\text{load}}/(R_c + R_{\text{load}})$;
predicted $\gamma_{\text{em}} = c/(R_c + R_{\text{load}})$